

WEEK 2 – DATA MODELING, DAX & SALES DASHBOARD

INTRODUCTION

The aim of this project is to create an interactive Superstore Sales Dashboard using Microsoft Power BI. The project focuses on data modeling, creating relationships between tables, developing DAX measures, and presenting data using different visualizations.

The Superstore dataset contains information about customers, products, orders, sales, profit, quantity, categories, and regions. The data was prepared and organized in Power BI before creating the dashboard.

The final dashboard provides a simple and interactive way to analyze sales performance, profit, orders, quantity, category-wise sales, and monthly sales. KPI cards, charts, slicers, and cross-filtering were used to make the dashboard easy to understand and explore.

OBJECTIVES

The main objectives of this project are:

- To import and understand the SampleSuperstore dataset.
- To clean and prepare the data using Power Query.
- To create supporting tables for better data organization.
- To establish relationships between tables.
- To create DAX measures for important calculations.
- To create a separate date table for time-based analysis.
- To design an interactive sales dashboard.
- To represent important information using KPI cards and charts.
- To implement slicers and cross-filtering.
- To analyze sales and profit performance using Power BI.

DATASET DESCRIPTION:

The **SampleSuperstore dataset** contains information about customers, products, orders, sales transactions, and business performance.

Important columns in the dataset include:

- Customer ID

- Customer Name
- Segment
- Product ID
- Product Name
- Category
- Sub-Category
- Sales
- Profit
- Quantity
- Order ID
- Order Date
- Region

The dataset was imported into Power BI and prepared for further analysis. The sales, profit, order, quantity, category, and date fields were mainly used to create the dashboard.

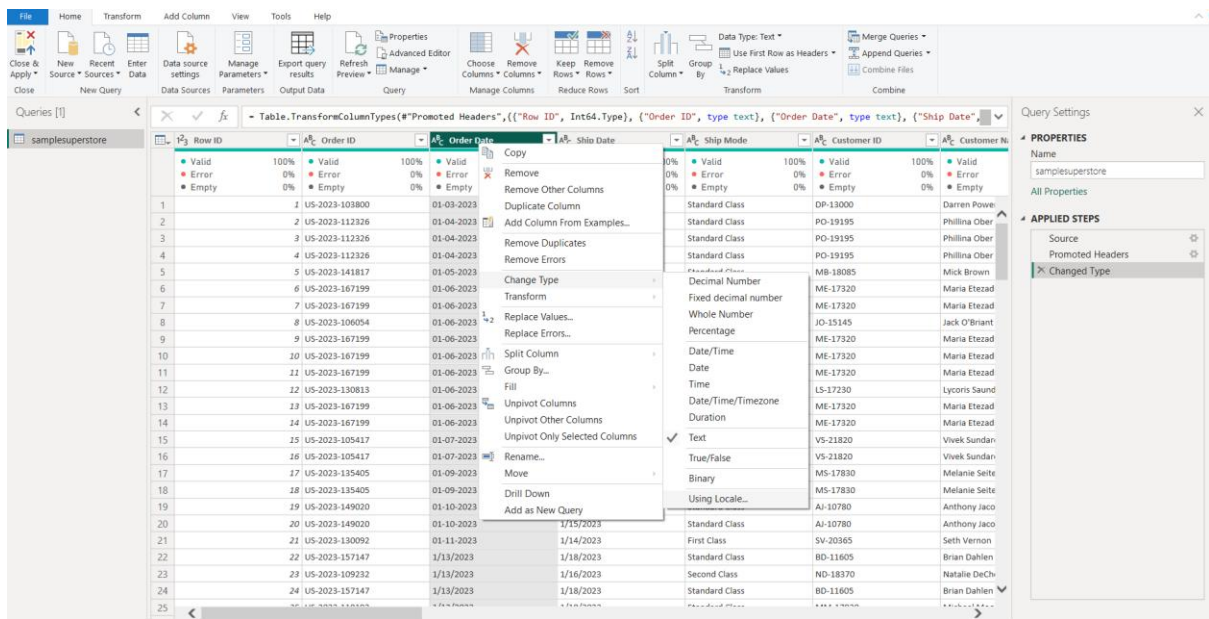
DATA PREPARATION:

The dataset was first imported into Power BI Desktop. Power Query was used to review and prepare the data before creating the data model.

The required columns and their data types were checked. The Order Date column was converted into the appropriate Date format. Text fields were cleaned and standardized where required.

Additional date-related information such as Year, Quarter, Month, and Day was created from the Order Date field to support time-based analysis.

These preparation steps helped to make the dataset consistent and suitable for creating visualizations and calculations.



DATA MODELING:

Data modeling was performed to organize the data and establish relationships between different tables.

The main sales table used in the project is:

SampleSuperstore

Supporting tables were created to improve the organization of the data.

DimCustomer

A DimCustomer table was created using DAX. It contains:

- Customer ID
- Customer Name
- Segment

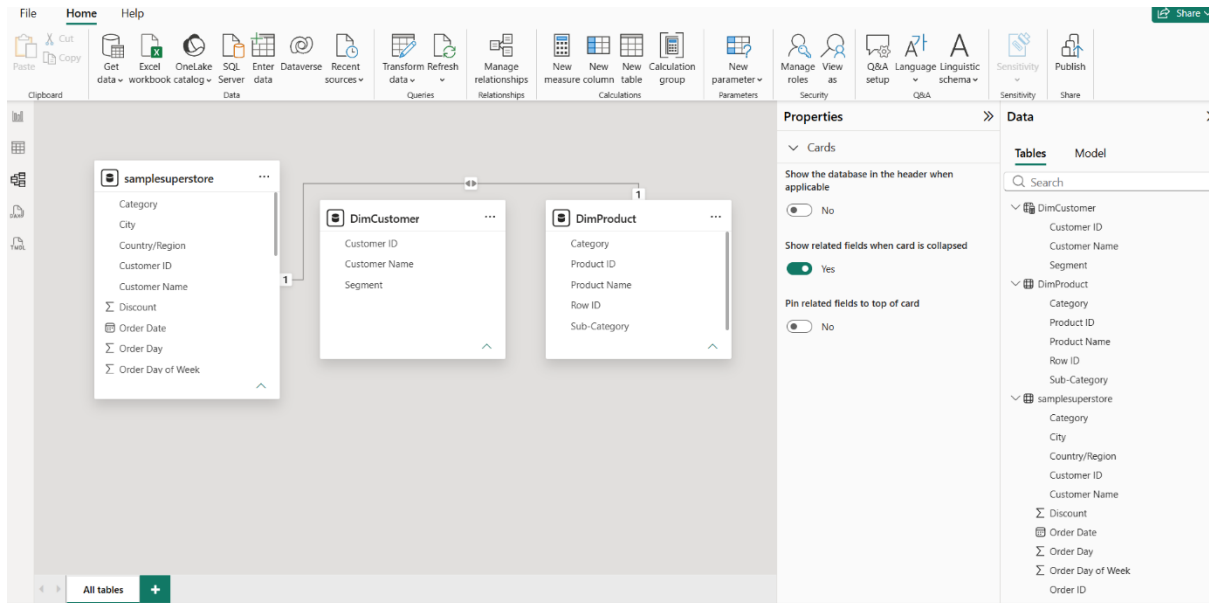
DimProduct

A DimProduct table was created using Power Query by referencing the SampleSuperstore table. The required product-related columns were retained, including:

- Product ID
- Product Name

- Category
- Sub-Category

The relationships between the supporting tables and the main sales table were created using the appropriate ID columns. Both relationships were configured as active one-to-many relationships.



CREATING THE DATE TABLE:

A separate date table named DimDate was created to support time-based analysis.

The table contains a continuous range of dates based on the minimum and maximum Order Date values in the SampleSuperstore table.

The following DAX expression was used:

```
DimDate =
    CALENDAR(
        MIN(SampleSuperstore[Order Date]),
        MAX(SampleSuperstore[Order Date])
    )
```

The DimDate table was then marked as a Date Table using its Date column.

This table helps Power BI perform date-based analysis and supports monthly and yearly sales analysis.

ESTABLISHING TABLE RELATIONSHIPS:

Relationships were created between the tables to build the Power BI data model.

The main relationships include:

DimCustomer[Customer ID] → SampleSuperstore[Customer ID]

DimProduct[Product ID] → SampleSuperstore[Product ID]

DimDate[Date] → SampleSuperstore[Order Date]

The relationships use:

- Cardinality: One-to-Many (1:*)
- Cross-filter direction: Single
- Active relationship: Yes

The dimension tables act as the one side, while the SampleSuperstore table acts as the many side because multiple sales records can belong to the same customer, product, or date.

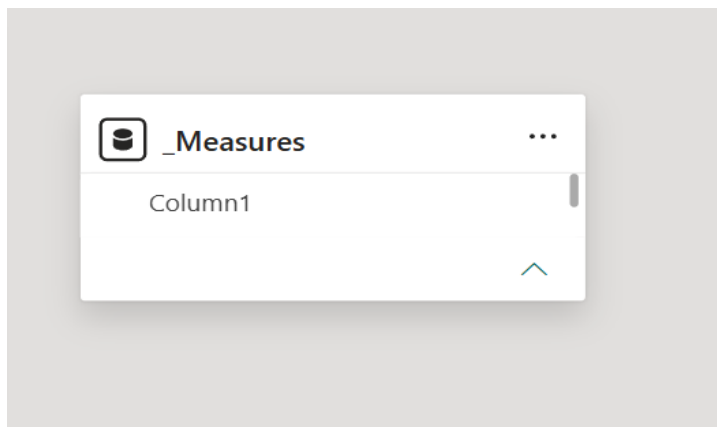
The relationships allow the supporting tables to filter the sales data correctly.

CREATING MEASURES TABLE:

A separate table named _Measures was created to store all the DAX measures used in the project.

Creating a separate measures table helps keep the calculations organized and makes the Power BI model easier to manage.

The important business calculations were created using the New Measure option in Power BI.



CREATING DAX MEASURES:

The following DAX measures were created for the dashboard.

1. Total Sales

Total Sales =

`SUM(SampleSuperstore[Sales])`

This measure calculates the total sales value from the SampleSuperstore table.

2. Total Profit

Total Profit =

`SUM(SampleSuperstore[Profit])`

This measure calculates the total profit.

3. Total Orders

Total Orders =

`DISTINCTCOUNT(SampleSuperstore[Order ID])`

This measure calculates the number of unique orders.

4. Total Quantity

Total Quantity =

`SUM(SampleSuperstore[Quantity])`

This measure calculates the total quantity of products sold.

5. Average Sales

Average Sales =

`AVERAGE(SampleSuperstore[Sales])`

This measure calculates the average sales value.

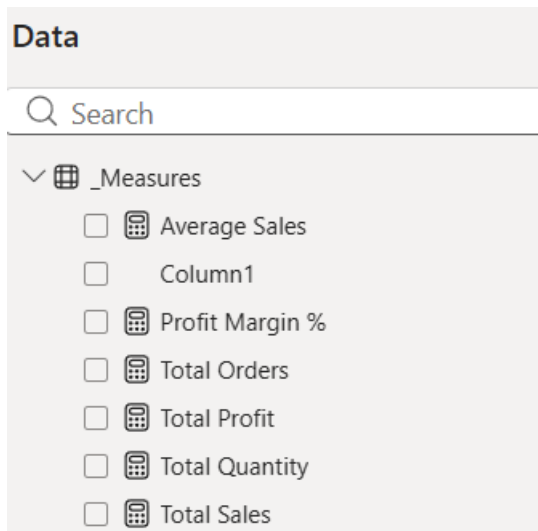
6. Profit Margin %

Profit Margin % =

`DIVIDE([Total Profit], [Total Sales], 0)`

This measure calculates the profit margin as a percentage of total sales.

These measures were used to display the main business metrics in the dashboard.



KPI CARDS:

KPI cards were used to display important business performance indicators in a simple format.

The dashboard contains six KPI cards:

1. Total Sales
2. Total Profit
3. Total Orders
4. Total Quantity
5. Average Sales
6. Profit Margin

The KPI cards provide a quick overview of the overall business performance and make the important values easy to understand.

KPI Values:

The dashboard displays approximately:

- Total Sales: 2.33M
- Total Profit: 292.30K
- Total Orders: 5K
- Total Quantity: 39K
- Average Sales: 228.23

- Profit Margin: 12.56%

These values are based on the SampleSuperstore dataset used in the project.

TOTAL SALES BY CATEGORY:

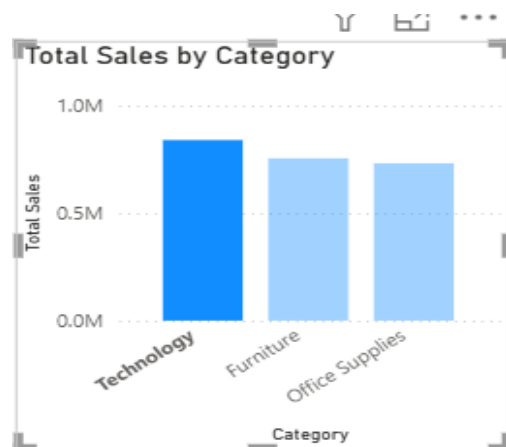
A column chart was created to compare total sales across different product categories.

The categories included are:

- Office Supplies
- Furniture
- Technology

The Total Sales measure was used as the value, while Category was used as the category field.

This visualization makes it easier to compare the sales performance of the different categories.



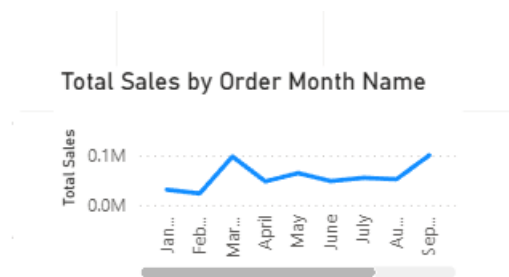
MONTHLY SALES ANALYSIS:

A line chart was created to analyze sales performance across different months.

The Order Month field was used on the X-axis and Total Sales was used on the Y-axis.

The month values were arranged in the correct January to December order using the Month Number column for sorting.

This visualization helps to understand how sales change over different months.

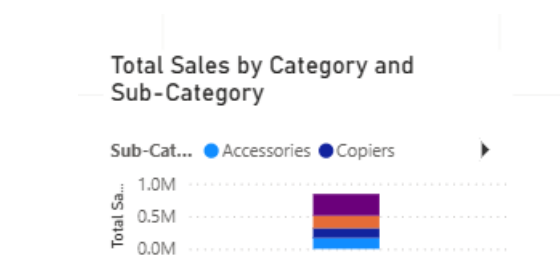


CATEGORY AND SUB-CATEGORY ANALYSIS:

A stacked column chart was created to analyze sales based on Category and Sub-Category.

Category provides the main grouping, while Sub-Category provides more detailed information about the products.

This visualization helps users examine the contribution of different sub-categories within each major category.



DECOMPOSITION TREE:

A Decomposition Tree was added to perform interactive analysis of Total Sales.

The Analyze field was set to:

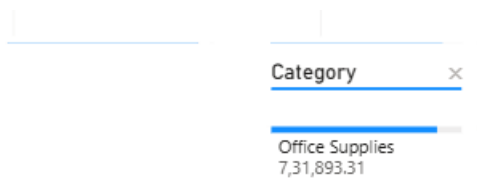
Total Sales

The Explain by fields include:

- Category
- Sub-Category
- Region
- Segment

The decomposition tree allows the Total Sales value to be broken down into different categories and levels. Users can select a category and continue analyzing the sales based on sub-category, region, and segment.

This provides a more detailed view of the factors contributing to total sales.



SLICER AND CROSS-FILTERING:

A Category Slicer was added to make the dashboard interactive.

The slicer contains:

- Office Supplies
- Furniture
- Technology

When a category is selected, the KPI cards and charts automatically update according to the selected category.

For example, selecting Technology displays the sales and other metrics related to the Technology category.

This demonstrates the cross-filtering functionality of Power BI and allows users to interactively explore the dataset. And the same for Region and Segment Slicer.

Category	Region	Segment
☐ Furniture	☐ Central	☐ Consumer
☐ Office Supplies	☐ East	☐ Corporate
☐ Technology	☐ South	☐ Home Office
	☐ West	

INTERACTIVE DASHBOARD:

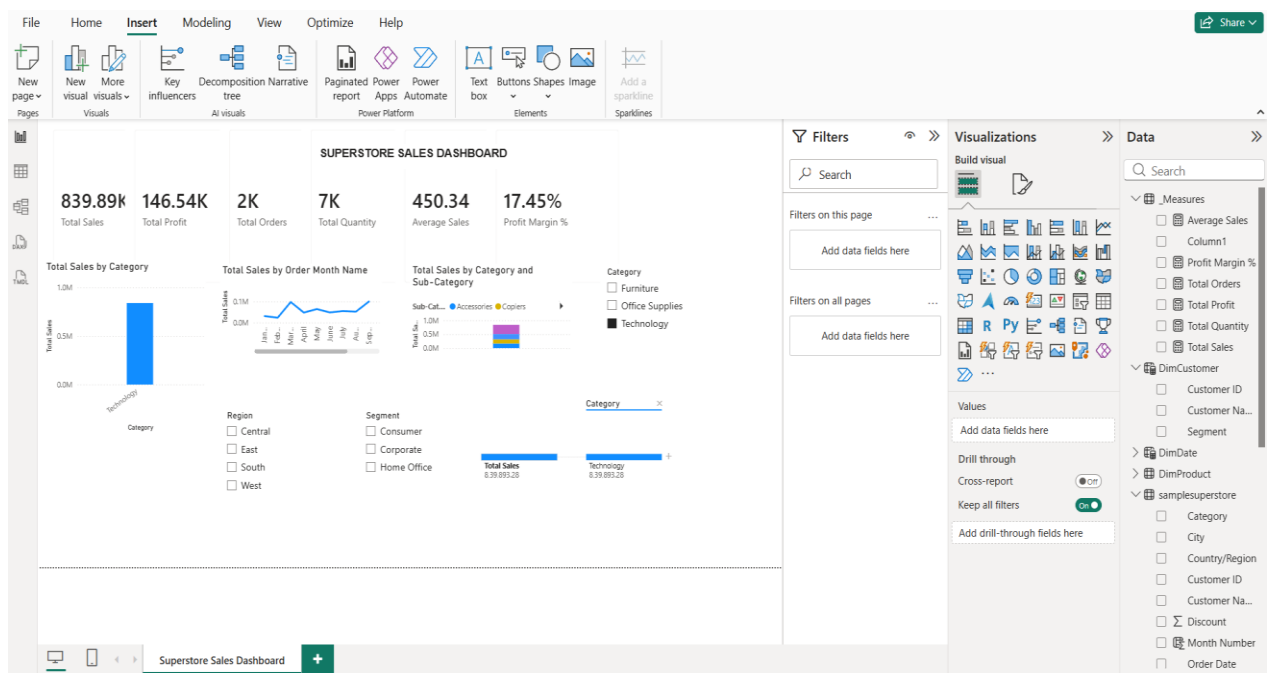
The final dashboard combines different Power BI visuals into a single interactive report page.

The dashboard contains:

- Six KPI cards
- Total Sales by Category chart

- Monthly Sales line chart
- Category and Sub-Category chart
- Decomposition Tree
- Category slicer
- Region slicer
- Segment slicer
- Drill-down functionality
- Cross-filtering

The dashboard was designed to provide important sales information in a simple and user-friendly format.



CONCLUSION:

The Week 2 Power BI project was successfully completed by creating a structured data model, establishing relationships, developing DAX measures, and designing an interactive sales dashboard.

The SampleSuperstore dataset was prepared and organized using Power Query. Supporting tables such as DimCustomer, DimProduct, and DimDate were created to improve data organization and analysis.

DAX measures were created to calculate Total Sales, Total Profit, Total Orders, Total Quantity, Average Sales, and Profit Margin. These measures were then used in KPI cards and different visualizations.

The use of charts, KPI cards, slicers, drill-down, and cross-filtering makes the dashboard interactive and easy to understand. The completed dashboard provides an effective way to analyze sales, profit, orders, quantity, category, sub-category, region, segment, and monthly performance using Power BI.

